

# Cross-Modal Sample Association and Dual-Source Adaptive Population Graph Learning for Multimodal Disease Prediction

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**Abstract.** Imaging data and non-imaging data provide complementary information for population-level disease prediction. However, existing population graph methods weakly connect cross-modal semantics, inadequately balance phenotype-derived constraints with data-driven adaptation, and underuse learned patient relationships. We propose an end-to-end framework comprising the Cross-modal Semantic Sample Association Module (CMSAA) for sample-level associations before graph construction, the Dual-source Adaptive Population Graph (DSAPG) for combining a phenotype-informed prior-constrained edge branch with dynamically learned semantic edge weights, and the Feature-Inductive Refinement Module (FIRM) for propagating high-level imaging and non-imaging embeddings under learned edge weights. The framework achieved 82.69% accuracy and an area under the receiver operating characteristic curve (AUC) of 89.04% on the ABIDE autism dataset, and 70.64% accuracy and 75.49% AUC on the Private Multimodal Coronary Artery Disease Dataset (PM-CAD). Ablations indicated contributions from all three components under the current protocol. Results indicate improved predictive performance.

**Keywords:** Multimodal learning · Population graph · Disease prediction · Graph neural network · Cross-modal association

## 1 Introduction

Advances in medical technology and data acquisition have produced large collections of heterogeneous multimodal biomedical data. Compared with a single modality, multimodal observations describe a patient’s condition from complementary perspectives and may support more comprehensive disease prediction [24, 25, 44]. They have therefore become increasingly important in computer-aided diagnosis systems.

This study considers patient population graph classification using imaging data and non-imaging data. For each patient, the imaging data may comprise resting-state functional connectivity, medical images, or another high-dimensional visual representation. The non-imaging data may include age, sex, clinical scales, symptoms, and other structured variables. The model learns discriminative node representations over a patient

population graph and performs binary disease prediction. We evaluate this formulation in two heterogeneous settings: autism prediction from imaging data and non-imaging data in the Autism Brain Imaging Data Exchange (ABIDE), and coronary artery disease (CAD) prediction from imaging data and non-imaging data in PM-CAD.

Existing multimodal disease prediction methods commonly integrate data sources through shared representations or feature fusion, including multiple-kernel learning [23, 41, 44], nonnegative matrix factorization [32, 49], and multi-branch deep networks [15, 28]. Shared representations help capture common information but may suppress modality-specific diagnostic cues. Feature concatenation and embedding-level fusion primarily model the modalities within each patient and pay less attention to structured relationships among patients [29, 38]. Effective multimodal patient modeling should therefore retain modality-specific information while exploiting inter-patient structure.

Graph neural networks (GNNs) provide a unified means of modeling these relationships [6, 16]. In a population graph, each subject is represented as a node, edges encode potential patient relationships, and neighborhood aggregation updates node representations [10, 19]. Population graphs can be constructed in advance using static similarity rules [17, 30] or learned and updated during training [26]. Static graphs conveniently incorporate clinical phenotype priors but cannot readily adapt to the task. Adaptive graphs offer greater flexibility, although their learned edges may not preserve clinical constraints. Prior studies have learned edge weights with pairwise encoders [14, 45] or used attention to estimate associations among non-imaging variables [37]. How to coordinate prior constraints and data-driven patient relationships nevertheless remains an open problem.

Three limitations motivate this work. First, semantic separation between imaging and non-imaging representations before graph construction can weaken sample-level association modeling. Mapping both modalities to the same numerical dimension does not, by itself, connect their high-level semantics. Second, a patient graph should balance constraints derived from non-imaging phenotypes against task-adaptive relationships learned from the data. A graph based entirely on fixed rules lacks flexibility, whereas a fully data-driven graph may discard clinically meaningful constraints. Third, after modality-specific graph encoding, existing approaches typically concatenate or attend to the encoded embeddings directly. They may therefore underuse the learned patient edge weights when forming the final predictive representation.

To address these limitations, we make the following contributions:

- We introduce a Cross-modal Semantic Sample Association Module (CMSAA), which models bidirectional sample-level associations before graph construction and augments the two modalities while retaining their original representations through residual paths.
- We propose a Dual-source Adaptive Population Graph (DSAF-PG). Its prior-constrained branch combines learned semantic similarity with a non-imaging affinity prior, while its dynamic branch learns task-related patient similarity. A learnable coefficient balances the two edge sources.
- We develop a Feature-Inductive Refinement Module (FIRM), which uses the learned patient edge weights to refine the high-level imaging and non-imaging embeddings

after graph encoding. The resulting embeddings are then integrated by the multi-modal attention mechanism inherited from MM-GTUNets.

## 2 Related Work

### 2.1 Multimodal Medical Representation Learning and Cross-Modal Alignment

Early multimodal disease prediction systems directly concatenated raw features for classification without explicitly modeling modality-distribution differences or redundancy in the expanded feature space [21, 33]. Subsequent methods fused modality-specific kernels [23, 41, 44], independently encoded and merged modality embeddings [15, 28], learned common representations through deep semi-nonnegative matrix factorization [49], or used bidirectional mappings to preserve modality information [27]. Because these approaches mainly fuse information at the input or embedding level, they provide limited modeling of cross-modal patient associations before graph construction. CMSAA addresses this gap and uses the enhanced representations for population-graph construction.

### 2.2 Population Graph Construction and Medical Graph Learning

GNNs are widely used for population-level multimodal disease prediction [11, 17, 38, 29]. Early methods used manually designed graphs, including kernel-based nonlinear fusion of modality-specific affinities [38], parallel graph convolutional branches [11], and population graphs defined by non-imaging data with imaging data as node features [17, 29]. Such prior-informed graphs offer a degree of interpretability, but their fixed topologies cannot adapt to changes in the data distribution.

Adaptive methods instead use attention or end-to-end structure learning. EV-GCN learns variational edge relationships [14], whereas MMGL jointly optimizes graph construction and representation learning [48]. MM-GTUNets combines an Affinity Metric Reward System (AMRS) with graph U-Net encoders, providing the non-imaging affinity estimation and multiscale graph backbone used here [3]. We extend it at three points: CMSAA adds sample-level cross-modal associations before graph construction, DSAF-PG combines prior-constrained and dynamic edge weights, and FIRM reuses patient edges after graph encoding. In DSAF-PG, AMRS affinity constrains rather than solely determines the propagation graph.

### 2.3 Post-Encoding Structural Refinement and Multimodal Fusion

Unlike convolutional neural networks on regular grids, patient graphs require topology-aware propagation [20, 42]; GNNs model such medical graphs through nodes and edges [19, 36, 43]. MMGL dynamically selects and integrates modality embeddings [48], whereas MS2-GNN ensembles structural and content representations [4]. These methods emphasize input or intermediate interactions, leaving post-encoding edge reuse underexplored. FIRM reuses learned patient edges to refine high-level modality representations before the inherited multimodal attention fusion.

### 3 Method

#### 3.1 Problem Formulation

Let the dataset be

$$\mathcal{D} = \left\{ \left( \mathbf{x}_i^{\text{img}}, \mathbf{x}_i^{\text{non}}, y_i \right) \right\}_{i=1}^N, \quad (1)$$

where  $\mathbf{x}_i^{\text{img}} \in \mathbb{R}^{d_1}$  and  $\mathbf{x}_i^{\text{non}} \in \mathbb{R}^{d_2}$  denote the feature vectors derived from the imaging data and non-imaging data of patient  $i$ , respectively, and  $y_i \in \{0, 1\}$  is the binary label. Stacking all samples gives

$$\mathbf{X}^{\text{img}} \in \mathbb{R}^{N \times d_1}, \quad \mathbf{X}^{\text{non}} \in \mathbb{R}^{N \times d_2}. \quad (2)$$

Each patient is a node in a population graph  $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathbf{A})$ , where  $\mathcal{V}$  is the node set,  $\mathcal{E}$  is the set of patient relationships, and  $\mathbf{A} \in \mathbb{R}^{N \times N}$  is a weighted adjacency matrix. The model learns a joint node representation  $\mathbf{H}^{\text{joint}}$  and predicts the label probabilities  $\hat{\mathbf{Y}}$ .

#### 3.2 Overall Architecture

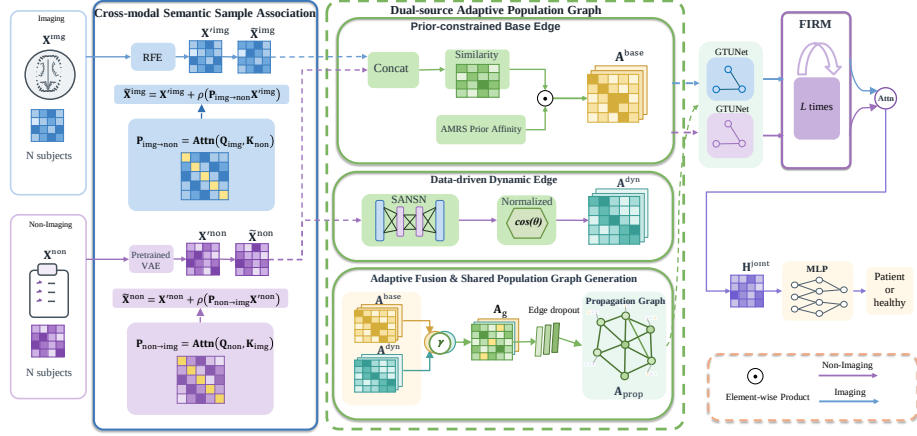
The proposed model builds on MM-GTUNets [3]. We retain its AMRS, GTUNet encoders, multimodal attention fusion, classifier, and modality-weighted graph regularization, and introduce CMSAA, DSAF-PG, and FIRM. These modules respectively enhance sample-level cross-modal associations before graph construction, fuse prior-constrained and data-driven edges, and refine the encoded features.

CMSAA first maps the imaging and non-imaging features to a common dimension and introduces cross-modal sample associations, producing  $\tilde{\mathbf{X}}^{\text{img}}$  and  $\tilde{\mathbf{X}}^{\text{non}}$ . DSAF-PG then uses the AMRS non-imaging affinity as a prior source, learns dynamic edges from the enhanced node features, and combines both branches into  $\mathbf{A}_g$ . Two structurally identical but independently parameterized GTUNet encoders extract multiscale representations of the two modalities over the same patient graph. FIRM subsequently refines both representations using the patient edge weights. Finally, the inherited multimodal attention module produces a joint embedding, and a multilayer perceptron predicts the class. Figure 1 summarizes the framework.

#### 3.3 Cross-Modal Semantic Sample Association Module

Multimodal learning requires more than matching feature dimensions; it must also address representation alignment, association learning, and information fusion [1]. Compressing heterogeneous modalities to the same dimension does not directly establish patient-level correspondence between their representation spaces. Directional cross-modal attention allows one modality to query another [39]. Although the original formulation targets multimodal language sequences, we adopt only its directional query–key matching principle for patient association estimation.

CMSAA constructs two  $N \times N$  cross-modal association matrices. Their weights arise from cross-modal query–key matches, but each matrix reweights patient features from



**Fig. 1.** Overall architecture. CMSAA aligns the imaging and non-imaging features and introduces cross-modal sample associations. DSAF-PG combines prior-constrained base edges with data-driven dynamic edges and applies edge dropout to form the propagation graph. Two GTUNet encoders learn modality-specific graph representations, which FIRM independently refines on the shared graph. The inherited attention module produces a joint embedding for disease prediction.

the same modality as the query. This design introduces cross-modally induced patient context without directly replacing modality-specific semantics. Residual connections preserve the initially aligned representation. Figure 2 illustrates the module.

The heterogeneous features are first mapped to dimension  $d$ . RFE selects disease-related imaging features:

$$\mathbf{X}'^{\text{img}} = \text{RFE}(\mathbf{X}^{\text{img}}; \theta_{\text{rfe}}) \in \mathbb{R}^{N \times d}. \quad (3)$$

The non-imaging data are encoded by a pretrained VAE:

$$\mathbf{X}'^{\text{non}} = \text{VAE}(\mathbf{X}^{\text{non}}; \theta_{\text{vae}}) \in \mathbb{R}^{N \times d}. \quad (4)$$

Independent linear projections generate the queries and keys:

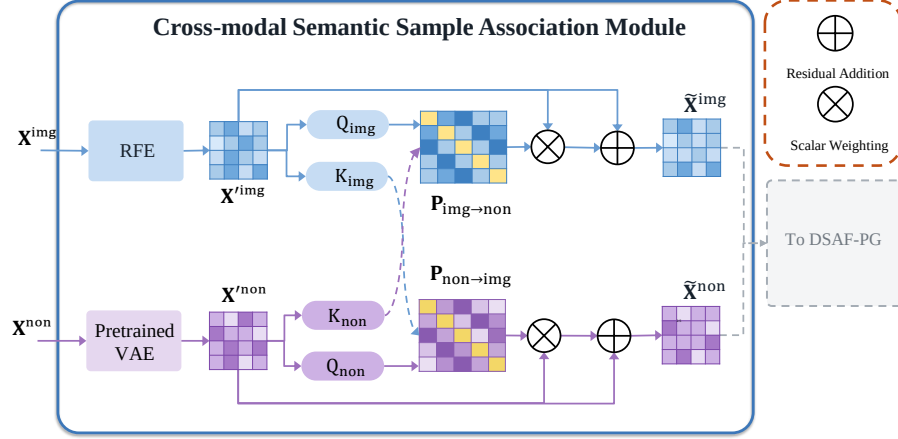
$$\mathbf{Q}_{\text{img}} = \mathbf{X}'^{\text{img}} \mathbf{W}_q^{\text{img}}, \quad \mathbf{K}_{\text{non}} = \mathbf{X}'^{\text{non}} \mathbf{W}_k^{\text{non}}, \quad (5)$$

$$\mathbf{Q}_{\text{non}} = \mathbf{X}'^{\text{non}} \mathbf{W}_q^{\text{non}}, \quad \mathbf{K}_{\text{img}} = \mathbf{X}'^{\text{img}} \mathbf{W}_k^{\text{img}}, \quad (6)$$

where all projection matrices belong to  $\mathbb{R}^{d \times d}$ . Row-wise scaled dot-product attention gives

$$\mathbf{P}_{\text{img} \rightarrow \text{non}} = \text{Softmax}\left(\frac{\mathbf{Q}_{\text{img}} \mathbf{K}_{\text{non}}^{\top}}{\sqrt{d}}\right), \quad (7)$$

$$\mathbf{P}_{\text{non} \rightarrow \text{img}} = \text{Softmax}\left(\frac{\mathbf{Q}_{\text{non}} \mathbf{K}_{\text{img}}^{\top}}{\sqrt{d}}\right), \quad (8)$$



**Fig. 2.** Cross-modal Semantic Sample Association Module. Recursive feature elimination (RFE) and a pretrained variational autoencoder (VAE) yield the initially aligned representations. Independent query and key projections generate bidirectional sample-level association matrices. These matrices guide same-modality patient-context aggregation, and residual addition produces the enhanced representations passed to DSAF-PG.

where both matrices are in  $\mathbb{R}^{N \times N}$ . Entry  $P_{\text{img} \rightarrow \text{non}}[i, j]$  measures the association induced between the imaging representation of patient  $i$  and the non-imaging representation of patient  $j$ ; the reverse matrix has the analogous interpretation.

The cross-modal weights guide same-modality patient-context reweighting:

$$\tilde{\mathbf{X}}_{\text{img}}^{\text{attn}} = \mathbf{P}_{\text{img} \rightarrow \text{non}} \mathbf{X}'^{\text{img}}, \quad \tilde{\mathbf{X}}^{\text{img}} = \rho \tilde{\mathbf{X}}_{\text{img}}^{\text{attn}} + \mathbf{X}'^{\text{img}}, \quad (9)$$

$$\tilde{\mathbf{X}}_{\text{non}}^{\text{attn}} = \mathbf{P}_{\text{non} \rightarrow \text{img}} \mathbf{X}'^{\text{non}}, \quad \tilde{\mathbf{X}}^{\text{non}} = \rho \tilde{\mathbf{X}}_{\text{non}}^{\text{attn}} + \mathbf{X}'^{\text{non}}. \quad (10)$$

Here  $\rho \in [0, 1]$  is the residual enhancement coefficient. A small  $\rho$  is used to preserve the individual feature distribution and limit excessive perturbation from cross-sample aggregation.

### 3.4 Dual-Source Adaptive Population Graph

The adjacency structure determines the scope of information propagation in a GNN. Real-world graphs may contain missing or noisy edges, and some datasets provide no usable topology. Jointly learning the graph structure and GNN parameters is therefore an established alternative to fixed construction rules [9]. In patient modeling, demographic attributes and laboratory measurements can also complement relationships inferred from latent representations [40]. Motivated by these observations, DSAF-PG combines a prior-constrained base graph and a dynamic semantic graph rather than relying exclusively on either fixed clinical rules or unconstrained learned relationships.

The CMSAA outputs are concatenated into a joint representation for pairwise estimation:

$$\widehat{\mathbf{X}} = \text{Concat}(\widetilde{\mathbf{X}}^{\text{img}}, \widetilde{\mathbf{X}}^{\text{non}}) \in \mathbb{R}^{N \times 2d}. \quad (11)$$

The  $i$ th row  $\widehat{\mathbf{x}}_i$  represents patient  $i$ .

**AMRS-Based Prior Affinity.** We retain the AMRS of MM-GTUNets [3] without changing its main structure. AMRS estimates the contributions of different variables in the non-imaging data and produces a prior affinity matrix  $\mathbf{C} \in \mathbb{R}^{N \times N}$ . For  $v$  variable categories, let

$$\boldsymbol{\alpha} = [\alpha_1, \dots, \alpha_v], \quad 0 < \alpha_u < 1, \quad \sum_{u=1}^v \alpha_u = 1. \quad (12)$$

The affinity between patients  $i$  and  $j$  is

$$C_{ij} = \text{Sigmoid} \left[ \sum_{u=1}^v \alpha_u \left( \beta_r^u R_{ij}^u + \beta_p^u P_{ij}^u + \beta_m^u M_{ij}^u \right) \right], \quad (13)$$

where  $R$ ,  $P$ , and  $M$  denote the reward, penalty, and motivation tables, respectively, and  $\{\beta_r^u, \beta_p^u, \beta_m^u\}$  are their coefficients. The sigmoid constrains  $C_{ij}$  to  $[0, 1]$ . Within DSAF-PG, AMRS is used only to generate the non-imaging prior affinity.

**Prior-Constrained Base Edges.** The AMRS affinity contains phenotype-derived relationships but does not include the multimodal semantics enhanced by CMSAA. We therefore multiply it by the semantic similarity of the joint representations:

$$A_{ij}^{\text{base}} = \text{SemSim}(\widehat{\mathbf{x}}_i, \widehat{\mathbf{x}}_j) C_{ij}. \quad (14)$$

The semantic similarity is an RBF kernel,

$$\text{SemSim}(\widehat{\mathbf{x}}_i, \widehat{\mathbf{x}}_j) = \exp \left[ -\frac{\delta(\widehat{\mathbf{x}}_i, \widehat{\mathbf{x}}_j)^2}{2\sigma^2} \right], \quad (15)$$

where  $\delta(\cdot, \cdot)$  is a feature distance and  $\sigma$  is the kernel bandwidth. Thus, the base edge is not a purely hand-crafted clinical prior; it combines learned multimodal similarity with the AMRS constraint.

**Dynamic Semantic Edge Learning.** To capture relationships not covered by the prior branch, a shared projection network  $f_\theta(\cdot)$  maps the enhanced non-imaging representation of each patient to a latent space:

$$\mathbf{z}_i = f_\theta(\widetilde{\mathbf{x}}_i^{\text{non}}), \quad \mathbf{z}_j = f_\theta(\widetilde{\mathbf{x}}_j^{\text{non}}). \quad (16)$$

The dynamic edge is a normalized cosine similarity,

$$A_{ij}^{\text{dyn}} = \frac{1}{2} \left[ \frac{\sum_{k=1}^{d_z} z_{ik} z_{jk}}{\sqrt{\sum_{k=1}^{d_z} z_{ik}^2} \sqrt{\sum_{k=1}^{d_z} z_{jk}^2}} + 1 \right], \quad (17)$$

where  $d_z$  is the latent dimension. The linear transformation maps cosine similarity to  $[0, 1]$ .

**Adaptive Dual-Source Fusion.** A learnable scalar controls the relative contribution of the two edge sources:

$$\gamma = \text{sigmoid}(\eta), \quad A_{g,ij} = \gamma A_{ij}^{\text{base}} + (1 - \gamma) A_{ij}^{\text{dyn}}, \quad (18)$$

where  $\eta$  is learned and  $\gamma \in [0, 1]$ . The complete patient adjacency matrix is

$$\mathbf{A}_g = [A_{g,ij}] \in \mathbb{R}^{N \times N}. \quad (19)$$

DropEdge randomly removes edges during training and acts as a structural perturbation while reducing message passing in each iteration [35]. We apply edge dropout only to  $\mathbf{A}_g$  during training to obtain the propagation matrix  $\mathbf{A}_{\text{prop}}$  used by both GTUNet and FIRM. At test time edge dropout is disabled and

$$\mathbf{A}_{\text{prop}} = \mathbf{A}_g. \quad (20)$$

Graph smoothness and degree regularization are always computed from the complete, undropped matrix  $\mathbf{A}_g$ .

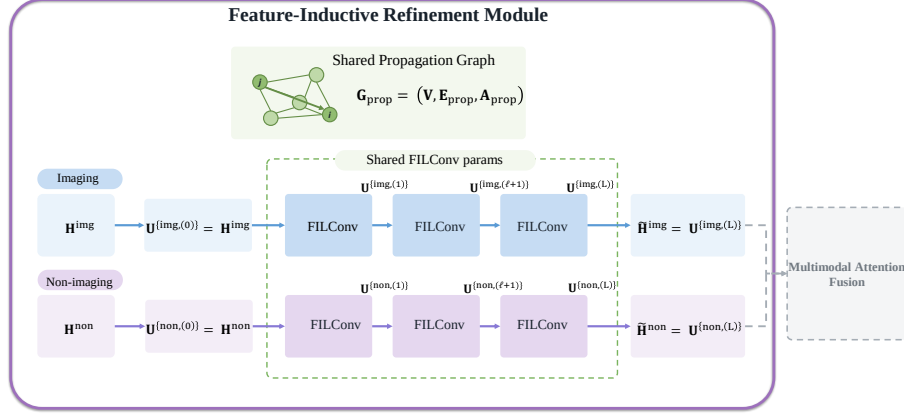
### 3.5 Graph Representation Learning and Multimodal Fusion

The DSAF-PG propagation graph contains both phenotype-informed constraints and data-driven relationships. Two GTUNet encoders independently process the imaging and non-imaging node features on this shared graph. FIRM then uses the same patient edges to refine their high-level embeddings. It does not transmit node features between modalities; cross-modal integration occurs only in the subsequent inherited attention module.

**Graph-Topology U-Net Encoder.** We reuse the GTUNetEncoder backbone from MM-GTUNets [3]. Define  $\mathcal{G}_{\text{prop}} = (\mathcal{V}, \mathcal{E}_{\text{prop}}, \mathbf{A}_{\text{prop}})$ , where  $\mathcal{E}_{\text{prop}}$  is the current propagation-edge set. For modality  $m \in \{\text{img}, \text{non}\}$ ,

$$\mathbf{H}^m = \text{GTUNet}_m(\tilde{\mathbf{X}}^m, \mathcal{G}_{\text{prop}}) \in \mathbb{R}^{N \times d_o}. \quad (21)$$

The default output dimension is  $d_o = 16$ . The two encoders have independent parameters but share the patient graph.



**Fig. 3.** Feature-Inductive Refinement Module. The imaging and non-imaging GTUNet embeddings are propagated independently over the same graph with shared FILConv parameters. After  $L$  layers, the refined representations are passed to multimodal attention fusion; no direct cross-modal message passing occurs inside FIRM.

**Feature-Inductive Refinement Module.** Neighborhood aggregation combines a node’s transformed features with information from its neighbors. GraphSAGE learns an aggregation function for this purpose [13]. APPNP further separates neural transformation from graph propagation [12], although it propagates prediction scores rather than high-level multimodal embeddings. FIRM is a lightweight post-encoder module that uses the continuous DSAF-PG weights to refine the GTUNet outputs.

As shown in Fig. 3, FIRM applies the same FILConv parameterization separately to both modalities. Parameter sharing keeps the refinement operators consistent, but the branches do not exchange node features. For  $m \in \{\text{img}, \text{non}\}$ , initialize

$$U^{m,(0)} = H^m \in \mathbb{R}^{N \times d_0}, \quad d_0 = d_o. \quad (22)$$

At layer  $\ell$ , node  $i$  is updated by

$$u_i^{m,(\ell+1)} = \sigma \left[ W_{\text{self}}^{(\ell)} u_i^{m,(\ell)} + \frac{1}{|\mathcal{N}(i)|} \sum_{j \in \mathcal{N}(i)} a_{ji}^{\text{prop}} W_{\text{neigh}}^{(\ell)} u_j^{m,(\ell)} \right], \quad (23)$$

for  $\ell = 0, \dots, L-1$ . Here  $\mathcal{N}(i)$  is the neighborhood of patient  $i$ ,  $a_{ji}^{\text{prop}}$  is the edge weight from  $j$  to  $i$ , and  $\sigma$  is ReLU. The shared matrices  $W_{\text{self}}^{(\ell)}$  and  $W_{\text{neigh}}^{(\ell)}$  respectively transform self and neighbor features. After  $L$  layers,

$$\tilde{H}^{\text{img}} = U^{\text{img},(L)}, \quad \tilde{H}^{\text{non}} = U^{\text{non},(L)}. \quad (24)$$

**Inherited Multimodal Attention Fusion.** The refined embeddings are integrated by the multimodal attention mechanism inherited from MM-GTUNets [3]. The shared

representation is

$$\tilde{\mathbf{H}}^{\text{sh}} = \frac{1}{2} \left( \tilde{\mathbf{H}}^{\text{img}} + \tilde{\mathbf{H}}^{\text{non}} \right). \quad (25)$$

The three attention representations are

$$\mathbf{T}^{\text{img}} = \tanh \left( \tilde{\mathbf{H}}^{\text{img}} \mathbf{W}_{\text{img}} + \mathbf{b}_{\text{img}} \right), \quad (26)$$

$$\mathbf{T}^{\text{non}} = \tanh \left( \tilde{\mathbf{H}}^{\text{non}} \mathbf{W}_{\text{non}} + \mathbf{b}_{\text{non}} \right), \quad (27)$$

$$\mathbf{T}^{\text{sh}} = \tanh \left( \tilde{\mathbf{H}}^{\text{sh}} \mathbf{W}_{\text{sh}} + \mathbf{b}_{\text{sh}} \right). \quad (28)$$

The final joint representation is

$$\begin{aligned} \mathbf{H}^{\text{joint}} &= \mathbf{T}^{\text{sh}} \odot \tilde{\mathbf{H}}^{\text{sh}} + \mathbf{T}^{\text{img}} \odot \tilde{\mathbf{H}}^{\text{img}} \\ &\quad + \mathbf{T}^{\text{non}} \odot \tilde{\mathbf{H}}^{\text{non}}, \end{aligned} \quad (29)$$

where  $\odot$  denotes element-wise multiplication.

Modality weights for graph regularization are derived from the energy ratio between each modality-specific attention representation and the shared representation:

$$s_m = \frac{\text{Tr} [\mathbf{T}^m (\mathbf{T}^m)^\top]}{\text{Tr} [\mathbf{T}^{\text{sh}} (\mathbf{T}^{\text{sh}})^\top] + \epsilon}, \quad m \in \{\text{img}, \text{non}\}, \quad (30)$$

$$[\omega_{\text{img}}, \omega_{\text{non}}] = \text{Softmax} ([s_{\text{img}}, s_{\text{non}}]). \quad (31)$$

These weights balance the modality-specific graph regularizers; they do not directly construct  $\mathbf{H}^{\text{joint}}$ .

### 3.6 Optimization Objective

A two-layer classifier produces the logits:

$$\mathbf{Z} = \text{ReLU} \left( \mathbf{H}^{\text{joint}} \mathbf{W}_1 + \mathbf{b}_1 \right), \quad (32)$$

$$\bar{\mathbf{Z}} = \text{BN}(\mathbf{Z}), \quad (33)$$

$$\mathbf{O} = \bar{\mathbf{Z}} \mathbf{W}_2 + \mathbf{b}_2, \quad \hat{\mathbf{Y}} = \text{Softmax}(\mathbf{O}). \quad (34)$$

Cross-entropy is applied to the logits. Let  $\mathcal{T}$  denote the training nodes, let  $K = 2$ , and let  $\mathbf{Y} \in \{0, 1\}^{N \times K}$  be the one-hot label matrix. Then

$$\mathcal{L}_{\text{ce}} = -\frac{1}{|\mathcal{T}|} \sum_{i \in \mathcal{T}} \sum_{k=1}^K Y_{ik} \log \hat{Y}_{ik}. \quad (35)$$

The graph regularizer of modality  $m$  is

$$\mathcal{L}_g^m = \lambda_s \mathcal{L}_{\text{smh}}^m + \lambda_d \mathcal{L}_{\text{deg}}, \quad m \in \{\text{img}, \text{non}\}. \quad (36)$$

Using the complete adjacency  $\mathbf{A}_g$ , define

$$\mathbf{D}_g = \text{diag}(\mathbf{A}_g \mathbf{1}), \quad \mathbf{L}_g = \mathbf{D}_g - \mathbf{A}_g. \quad (37)$$

The smoothness term is

$$\mathcal{L}_{\text{smh}}^m = \frac{1}{N^2} \text{Tr} \left[ (\tilde{\mathbf{H}}^m)^\top \mathbf{L}_g \tilde{\mathbf{H}}^m \right], \quad (38)$$

which penalizes edge-weighted representation differences between connected patients. The degree regularizer is

$$\mathcal{L}_{\text{deg}} = -\frac{1}{N} \mathbf{1}^\top \log (\mathbf{A}_g \mathbf{1} + \epsilon), \quad (39)$$

which discourages node degrees from collapsing toward zero. If  $V$  denotes the AMRS state-action value, the reward regularizer is

$$\mathcal{L}_r = \frac{1}{V + \epsilon}. \quad (40)$$

The total objective is

$$\begin{aligned} \mathcal{L}_{\text{total}} = & \mathcal{L}_{\text{ce}} + \omega_{\text{img}} \left( \lambda_s \mathcal{L}_{\text{smh}}^{\text{img}} + \lambda_d \mathcal{L}_{\text{deg}} \right) \\ & + \omega_{\text{non}} \left( \lambda_s \mathcal{L}_{\text{smh}}^{\text{non}} + \lambda_d \mathcal{L}_{\text{deg}} + \lambda_r \mathcal{L}_r \right), \end{aligned} \quad (41)$$

with default coefficients

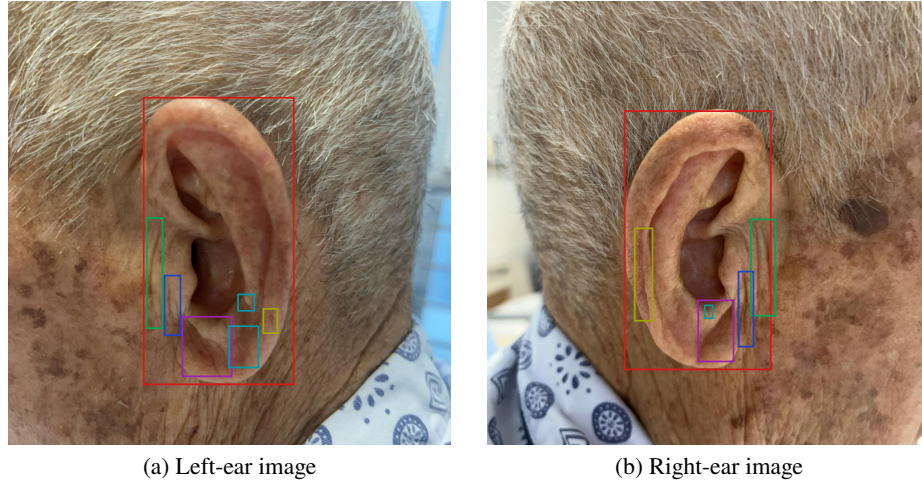
$$\lambda_s = 1, \quad \lambda_d = 10^{-4}, \quad \lambda_r = 10^{-2}. \quad (42)$$

## 4 Experiments and Results

### 4.1 Datasets and Preprocessing

**PM-CAD.** PM-CAD contains paired imaging data and non-imaging data collected from 196 participants at Tongji Hospital, Wuhan, between December 1, 2024 and December 7, 2025: 148 healthy controls (HCs) and 48 patients with coronary atherosclerotic heart disease (CAD). The task is binary CAD-versus-HC classification.

The construction of PM-CAD involved participant recruitment, clinical questionnaire administration, bilateral ear-image acquisition, auricular crease annotation, image-feature extraction, and data quality control. Participants with CAD and healthy controls were recruited at the hospital. During their clinical visits or examinations, structured questionnaires were administered to collect symptom-related information, including chest pain and chest tightness, as well as clinical risk factors such as hypertension, diabetes, dyslipidemia, and smoking history. Left- and right-ear images were then acquired separately using a standardized imaging procedure to ensure that the auricular crease regions were complete, clearly visible, and free from occlusion. After image acquisition, the visible longitudinal auricular crease regions in both ears were manually annotated. To



**Fig. 4.** De-identified bilateral ear images with manually annotated longitudinal auricular crease regions for a representative participant from PM-CAD. Bounding boxes indicate manually identified crease regions. Potentially identifiable information was masked to protect participant privacy.

illustrate the ear images and the annotation procedure, Fig. 4 presents the de-identified left- and right-ear images of a representative participant. The annotation boxes indicate the manually identified longitudinal auricular crease regions of interest (ROIs). Potentially identifiable information was masked, and only the ear regions required for the present analysis were retained. Multiple longitudinal crease ROIs were subsequently cropped from each ear image, and a pretrained CLIP ViT-B/32 image encoder was used to extract a 512-dimensional deep visual feature vector from each ROI. For each ear, the feature vectors of all annotated ROIs were averaged to produce a 512-dimensional ear-level representation. The left- and right-ear representations were then concatenated to form a 1,024-dimensional auricular-image feature vector. Finally, the image features, clinical questionnaire variables, and diagnostic labels were linked using anonymized participant identifiers to construct the final PM-CAD multimodal cohort.

Non-imaging data comprise age, sex, chest tightness, tenderness, pain location, extent, nature, activity, severity, response, duration, relief with rest, restlessness, pallor, sweating, dyspnea, cold extremities, lower-limb weakness, syncope, transient visual blackout, cough and sputum, nausea, vomiting, dyslipidemia, hypertension, diabetes, family history, smoking and alcohol consumption, diet, physical activity, sleep status, anxiety history, and regular physical examinations. Age remained a continuous floating-point variable; categorical variables were ordinally encoded, with missingness as a separate category before concatenation into  $\mathbf{x}^{\text{non}}$ .

**ABIDE Dataset.** ABIDE I contains imaging data from more than 1,000 resting-state functional magnetic resonance imaging (rs-fMRI) scans and corresponding non-imaging data collected across 24 sites [8]. Following MM-GTUNets [3], we used 871 subjects: 468 HCs and 403 participants with autism spectrum disorder (ASD).

**Table 1.** Demographic characteristics of ABIDE and the PM-CAD Dataset. Values for age are mean  $\pm$  standard deviation.

| Dataset | Diagnosis | Subjects | Female/Male | Age (years)       |
|---------|-----------|----------|-------------|-------------------|
| ABIDE   | HC        | 468      | 90/378      | $16.84 \pm 7.23$  |
| ABIDE   | ASD       | 403      | 54/349      | $17.07 \pm 7.95$  |
| PM-CAD  | HC        | 148      | 56/92       | $61.71 \pm 10.40$ |
| PM-CAD  | CAD       | 48       | 11/37       | $62.32 \pm 11.15$ |

The rs-fMRI preprocessing followed the general procedure of Parisot et al. [30]. Data were processed with configurations based on the Configurable Pipeline for the Analysis of Connectomes (C-PAC) [7] and Athena [2]. The brain was divided into 116 regions of interest with an automated anatomical labeling atlas [34]. The mean time series of each region was extracted, and Pearson correlation coefficients formed a functional connectivity matrix for each subject. The upper-triangular elements were vectorized to produce  $\mathbf{x}^{\text{img}}$ . Categorical phenotypes were ordinally encoded, numerical variables were converted to floating-point values, and the resulting variables were concatenated into  $\mathbf{x}^{\text{non}}$ .

## 4.2 Experimental Protocol

The proposed method was evaluated using stratified ten-fold cross-validation on ABIDE (HC versus ASD) and PM-CAD (HC versus CAD). In each round, eight folds were used for training, one fold for validation, and one fold for testing, with the folds rotated so that each fold served as the test fold once.

The evaluation follows a transductive setting. Node features from the training, validation, and test folds were available during population graph construction and graph-based message passing. Class labels from the validation and test folds were excluded from both population graph construction and supervised parameter optimization. The classification loss was computed only on the training nodes. The validation fold was used for model selection and early stopping, whereas the test labels were accessed only for final performance evaluation.

Accuracy (ACC), sensitivity (SEN), specificity (SPE), and the area under the receiver operating characteristic curve (AUC) were computed on the test fold. Tables report the mean and standard deviation across the ten test folds.

## 4.3 Implementation Details

The model was implemented in PyTorch and optimized with Adam [18] on a server equipped with four NVIDIA GeForce RTX 3090 GPUs. During initial VAE pretraining, the learning rate was  $10^{-3}$ , weight decay was  $5 \times 10^{-4}$ , and training continued for 3,000 epochs. After the pretrained VAE was frozen, all remaining model parameters were optimized for 300 epochs with a learning rate of  $10^{-4}$  and weight decay of  $5 \times 10^{-4}$ . Both feature dropout and edge dropout were set to 0.3. Early stopping used a patience of 100 epochs.

For feature alignment, RFE was fitted using the imaging features and corresponding labels from the training folds in each cross-validation round. The fitted feature selector was then applied unchanged to the training, validation, and test nodes.

Under the transductive evaluation setting, the VAE was pretrained in an unsupervised manner using the non-imaging features of all nodes in the current population graph, including the training, validation, and test nodes. No class labels were used during VAE pretraining. After pretraining, the VAE parameters were frozen before optimization of the remaining model components.

#### 4.4 Baselines and Evaluation Metrics

We compared the proposed method with support vector machines (SVM), multilayer perceptrons (MLP), and recent graph-based disease prediction methods: BrainGNN [22], DGCN [47], AL-NEGAT [5], A-GCL [46], Pop-GCN [30], GATE [31], EV-GCN [14], MMGL [48], and MM-GTUNets [3]. Evaluation used ACC, SEN, SPE, and AUC, all expressed as percentages.

#### 4.5 Main Results

Table 2 compares the proposed method with traditional machine-learning, graph-structured, and population-graph-prior methods, denoted by the prefixes T, B, and P, respectively.

Based on Table 2, three observations can be made. First, the strongest baselines are predominantly graph- or population-graph-based, although their rankings vary across datasets. On ABIDE, MM-GTUNets provides the highest baseline ACC of 81.30% and AUC of 88.72%, whereas on PM-CAD, P-MMGL provides the highest baseline ACC of 67.71% and AUC of 71.54%. This variation reflects differences between the two cohorts.

Second, the proposed method achieves the highest ACC and AUC on both datasets. On ABIDE, it attains an ACC of 82.69%, an AUC of 89.04%, and the highest SEN of 90.19%. On PM-CAD, it reaches an ACC of 70.64% and an AUC of 75.49%, exceeding the strongest baseline results for both metrics. The larger gains on PM-CAD suggest that the proposed cross-modal association and adaptive graph-learning strategy may be particularly beneficial for this cohort.

Third, the gains are not uniform across all metrics. A-GCL achieves the highest SPE of 80.95% on ABIDE, while P-MMGL and MM-GTUNets achieve the highest SEN of 74.62% and SPE of 68.33%, respectively, on PM-CAD. Relative to MM-GTUNets, the proposed method increases SEN but reduces SPE on both datasets, indicating a sensitivity–specificity trade-off.

#### 4.6 Ablation Study

Table 3 evaluates the incremental and removal-based contributions of CMSAA, DSAF-PG, and FIRM. The baseline is MM-GTUNets.

**Table 2.** Classification results on ABIDE and the PM-CAD Dataset. Values are mean (standard deviation), in percent. The best result in each column is shown in bold.

| ABIDE           |                     |                     |                     |                     |
|-----------------|---------------------|---------------------|---------------------|---------------------|
| Method          | ACC                 | SEN                 | SPE                 | AUC                 |
| T-SVM           | 66.02 (0.35)        | 65.34 (0.26)        | 78.83 (0.43)        | 64.97 (0.37)        |
| T-MLP           | 72.69 (0.84)        | 72.02 (2.84)        | 73.45 (0.61)        | 77.69 (1.02)        |
| B-BrainGNN      | 69.76 (3.80)        | 67.47 (3.10)        | 73.28 (3.26)        | 72.50 (3.10)        |
| B-DGCN          | 71.83 (2.90)        | 70.90 (2.59)        | 71.80 (3.10)        | 72.45 (2.98)        |
| B-AL-NEGAT      | 73.17 (3.32)        | 78.18 (2.66)        | 73.28 (3.14)        | 75.02 (2.56)        |
| B-A-GCL         | 79.04 (2.40)        | 81.42 (3.03)        | <b>80.95 (3.19)</b> | 82.86 (2.91)        |
| P-Pop-GCN       | 68.43 (0.86)        | 78.32 (0.89)        | 57.51 (5.99)        | 73.90 (3.08)        |
| P-GATE          | 74.65 (2.50)        | 75.59 (2.43)        | 76.87 (2.27)        | 76.87 (2.27)        |
| P-EV-GCN        | 80.95 (0.27)        | 83.74 (0.61)        | 77.69 (0.18)        | 82.37 (0.29)        |
| P-MMGL          | 80.34 (0.18)        | 80.25 (0.21)        | 77.16 (3.15)        | 79.83 (1.06)        |
| P-MM-GTUNets    | 81.30 (0.54)        | 82.93 (0.87)        | 79.43 (0.61)        | 88.72 (0.48)        |
| Proposed method | <b>82.69 (0.81)</b> | <b>90.19 (0.74)</b> | 74.04 (3.50)        | <b>89.04 (0.52)</b> |

| PM-CAD Dataset  |                     |                     |                     |                     |
|-----------------|---------------------|---------------------|---------------------|---------------------|
| Method          | ACC                 | SEN                 | SPE                 | AUC                 |
| T-SVM           | 58.50 (1.80)        | 64.46 (2.40)        | 54.18 (1.20)        | 65.76 (3.10)        |
| T-MLP           | 63.53 (1.60)        | 72.26 (2.10)        | 56.47 (1.80)        | 68.51 (2.70)        |
| B-BrainGNN      | 66.05 (2.50)        | 68.15 (3.20)        | 53.36 (5.80)        | 65.07 (3.50)        |
| B-DGCN          | 65.47 (2.20)        | 70.44 (2.80)        | 55.63 (5.30)        | 67.83 (3.10)        |
| B-AL-NEGAT      | 60.21 (2.10)        | 72.82 (2.60)        | 57.25 (4.90)        | 68.50 (2.90)        |
| B-A-GCL         | 63.54 (1.90)        | 74.36 (2.40)        | 58.18 (4.60)        | 70.21 (2.60)        |
| P-Pop-GCN       | 64.19 (2.30)        | 69.42 (3.00)        | 52.46 (1.90)        | 66.31 (3.20)        |
| P-GATE          | 66.63 (2.00)        | 72.06 (2.50)        | 56.39 (3.00)        | 68.27 (2.80)        |
| P-EV-GCN        | 66.27 (1.70)        | 74.56 (2.20)        | 60.52 (4.30)        | 71.36 (2.40)        |
| P-MMGL          | 67.71 (1.80)        | <b>74.62 (2.30)</b> | 60.14 (4.40)        | 71.54 (2.50)        |
| P-MM-GTUNets    | 66.67 (1.27)        | 66.18 (1.37)        | <b>68.33 (4.14)</b> | 69.99 (3.55)        |
| Proposed method | <b>70.64 (1.13)</b> | 72.70 (1.41)        | 61.66 (4.52)        | <b>75.49 (3.23)</b> |

Adding CMSAA to the baseline improved performance on both datasets. Adding DSAF-PG produced a further AUC increase on PM-CAD, suggesting that the dual-source edge design may support patient-relationship modeling in the present task. The full model attained the highest AUC after FIRM was included, indicating an incremental contribution from edge-guided post-encoding refinement under this model and evaluation protocol.

The removal comparisons lead to the same cautious interpretation. On PM-CAD, removing CMSAA reduced AUC from 75.49% to 73.60%, removing DSAF-PG reduced it to 72.40%, and removing FIRM reduced it to 74.10%. The three modules form a continuous pipeline of cross-modal sample association, dual-source graph construction, and post-encoding refinement.

#### 4.7 Modality Contribution

Table 4 compares inputs based on non-imaging data alone, imaging data alone, and both data modalities.

On PM-CAD, non-imaging data alone provided some discrimination, indicating predictive associations between certain clinical variables and CAD labels. Imaging data alone performed better, suggesting that the imaging representation contains task-related information for CAD classification under the current dataset and split protocol.

**Table 3.** Module ablation results. Values are mean (standard deviation), in percent.

| Variant                    | CMSAA | DSAF-PG | FIRM | ABIDE               |                     | PM-CAD Dataset      |                     |
|----------------------------|-------|---------|------|---------------------|---------------------|---------------------|---------------------|
|                            |       |         |      | ACC                 | AUC                 | ACC                 | AUC                 |
| Baseline                   | –     | –       | –    | 81.30 (0.54)        | 88.72 (0.48)        | 66.67 (1.27)        | 69.99 (3.55)        |
| Baseline + CMSAA           | ✓     | –       | –    | 81.72 (0.62)        | 88.83 (0.55)        | 67.86 (1.55)        | 71.20 (3.20)        |
| Baseline + DSAF-PG         | –     | ✓       | –    | 82.03 (0.70)        | 88.88 (0.50)        | 68.45 (1.70)        | 72.85 (2.90)        |
| Baseline + FIRM            | –     | –       | ✓    | 81.58 (0.65)        | 88.79 (0.55)        | 67.45 (1.60)        | 71.05 (3.10)        |
| Baseline + CMSAA + DSAF-PG | ✓     | ✓       | –    | 82.31 (0.76)        | 88.96 (0.54)        | 69.62 (1.45)        | 74.10 (2.35)        |
| Baseline + DSAF-PG + FIRM  | –     | ✓       | ✓    | 82.12 (0.73)        | 88.91 (0.58)        | 69.05 (1.62)        | 73.60 (2.75)        |
| Baseline + CMSAA + FIRM    | ✓     | –       | ✓    | 81.86 (0.66)        | 88.80 (0.60)        | 68.36 (1.80)        | 72.40 (3.05)        |
| Full model                 | ✓     | ✓       | ✓    | <b>82.69 (0.81)</b> | <b>89.04 (0.52)</b> | <b>70.64 (1.13)</b> | <b>75.49 (3.23)</b> |

**Table 4.** Contribution of the input modalities. Values are mean (standard deviation), in percent.

| Input                           | ABIDE               |                     | PM-CAD Dataset      |                     |
|---------------------------------|---------------------|---------------------|---------------------|---------------------|
|                                 | ACC                 | AUC                 | ACC                 | AUC                 |
| Non-imaging data                | 52.00 (0.24)        | 50.28 (0.38)        | 64.29 (1.85)        | 68.70 (3.10)        |
| Imaging data                    | 76.49 (0.80)        | 85.22 (0.72)        | 68.45 (1.62)        | 73.25 (2.95)        |
| Imaging data + non-imaging data | <b>82.69 (0.81)</b> | <b>89.04 (0.52)</b> | <b>70.64 (1.13)</b> | <b>75.49 (3.23)</b> |

Combining imaging data and non-imaging data yielded the highest AUC and therefore supports predictive complementarity in this internal cohort.

## 5 Discussion

### 5.1 Performance and the Sensitivity–Specificity Trade-Off

The proposed framework achieved strong AUC values on both datasets and showed a more visible improvement over MM-GTUNets on PM-CAD. Both datasets, however, exhibited increased sensitivity and decreased specificity. The model may therefore favor positive-case detection. High sensitivity can be valuable in screening, whereas reduced specificity produces additional false positives. Threshold calibration, class-imbalance strategies, and cost-sensitive learning should be investigated to obtain application-dependent operating points.

### 5.2 Limitations

This study has several limitations. First, the experiments address binary classification with complete paired modalities. PM-CAD is small and single-center, and generalization to multicenter, multiclass, and missing-modality settings remains unknown. Second, CMSAA and DSAF-PG construct  $N \times N$  sample-relation matrices, increasing computation and memory requirements for larger cohorts. A new patient must also be related to the reference cohort; sparse graph construction and inductive inference are therefore important future directions. Third, the present framework integrates only imaging data and non-imaging data. It could be extended to genomic and other data sources, as well as disease staging, risk prediction, and prognosis. Finally, the source manuscript reports fold-level means and standard deviations but no paired significance test, confidence interval, external validation, or detailed control of site and acquisition confounders.

## 6 Conclusion

We proposed a multimodal population graph framework that combines cross-modal sample association with dual-source adaptive graph learning. CMSAA introduces sample-level cross-modal associations before graph construction. DSAF-PG combines a prior-constrained base edge with a data-driven dynamic edge, and FIRM performs edge-guided refinement of the modality-specific embeddings after graph encoding. The inherited MM-GTUNets attention mechanism then produces the joint representation. The framework achieved competitive predictive performance on ABIDE and PM-CAD, and the ablations indicate contributions from all three proposed modules under the current evaluation protocol.

On PM-CAD, combining imaging data and non-imaging data produced the highest AUC, supporting predictive complementarity under the present internal validation. It does not establish a stable medical mechanism for CAD. Future work should expand the cohort, perform multicenter external validation, add fold-level statistical comparisons, analyze cross-modal representation consistency and graph stability, inspect learned edges, compare alternative fusion and refinement operators, and evaluate clinical confounding and interpretability.

## Code and Data Availability

The source code, model configurations, and data preprocessing scripts are publicly available at: <https://github.com/baiXianShen136/CSDA-PGNet>.

The ABIDE dataset used in this study is publicly available through its official repository, subject to the corresponding data-use agreement. The private PM-CAD dataset cannot be publicly released because of participant privacy, ethical requirements, and institutional data-use restrictions. Non-sensitive materials, including data format descriptions, configuration files, preprocessing scripts, and synthetic examples, are provided in the code repository.

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